

Field-Centric Driving Practice

Two-Hour Full-Field Session

Purpose

Build confidence with field-centric mecanum controls, maintain consistent field-relative movement from any robot heading, and develop fast, repeatable match cycles.

In field-centric control, joystick directions stay aligned with the field: pushing away from the driver moves the robot away from the driver even after it rotates. Rotation remains controlled separately.

Equipment and Setup

- Competition robot, charged batteries, Driver Station, and gamepads
- Full field with normal obstacles and scoring structures
- 8–12 cones, game pieces, or floor markers
- Tape for lanes, stopping boxes, and a visible field-direction reference
- Stopwatch and score sheet

Mark field-forward where drivers can see it. Inspect the robot, confirm configured forward and heading reset, and agree on an emergency-stop call. Nobody enters the field until the robot is disabled.

Session Schedule

Time	Exercise	Primary focus
0:00–0:10	Setup, heading, and reset check	Safety and coordinate accuracy
0:10–0:25	Field-direction warm-up	Field-relative movement
0:25–0:45	Fixed-heading shapes	Translation independent of heading
0:45–1:05	Move-and-rotate course	Independent translation and rotation
1:05–1:25	Target-facing approaches	Useful heading while traveling
1:25–1:43	Full-field routes	Speed and repeatability
1:43–1:55	Heading recovery drill	Recognizing and correcting drift
1:55–2:00	Timed challenge and review	Performance baseline

0:00–0:10 — Setup, Heading, and Reset Check

1. Inspect wheels, hubs, mechanisms, battery, and wiring.
2. Clear the field and mark forward, backward, left, and right.
3. Place the robot in the standard starting orientation and initialize.
4. Verify field-forward, then test all four directions at headings of 0, 90, 180, and 270 degrees.
5. Practice the team's exact heading-reset procedure twice.
6. Each driver completes one slow field-relative perimeter lap.

Do not continue if field directions are inconsistent. Correct the heading reference first.

0:10–0:25 — Field-Direction Warm-Up

1. The coach calls a cardinal field direction.
2. The driver moves one robot length without rotating.
3. Rotate 45–90 degrees and repeat.
4. Move to each wall from several robot headings.
5. Finish with diagonal direction calls.

Look at the field and destination instead of converting directions from the robot's front.

Success target: 12 correct initial movements in a row from at least four headings.

0:25–0:45 — Fixed-Heading Shapes

Mark a large square, diamond, and figure eight.

1. Drive the square while pointing toward the audience side.
2. Repeat while pointing toward the scoring side.
3. Drive the diamond without changing heading.
4. Drive the figure eight while holding a fixed heading.
5. Repeat at slow and medium speed.

Watch for unintended rotation, curved paths, and abrupt corrections.

Success target: three clean shapes with no contact and under 10 degrees of heading change.

0:45–1:05 — Move-and-Rotate Course

1. Strafe field-right while slowly rotating one full turn.
2. Drive field-forward while rotating 180 degrees; stop in a box.
3. Slalom through cones while continuously facing the center cone.
4. Circle a marker while keeping the robot pointed toward it.
5. Repeat in the opposite direction.

Begin with small rotation inputs. Translation should remain field-relative throughout the turn.

Success target: three clean runs with no wrong-direction start or missed stopping box.

1:05–1:25 — Target-Facing Approaches

Mark realistic scoring, pickup, or interaction positions. For each target:

1. Start from center or a collection area with a random heading.
2. Drive toward the target using field-relative translation.
3. Rotate during travel to reach the required final heading.
4. Stop in alignment and complete the action when possible.
5. Back away in a chosen field direction while watching traffic.

Practice facing the target, facing away for a rear mechanism, and holding a sideways orientation. Prefer one smooth arc over separate translate-and-rotate steps.

Success target: 8 of 10 alignments with at most one final correction.

1:25–1:43 — Full-Field Routes

Start → near pickup → far scoring area → center obstacle → near scoring area → park

1. Learn at 50% speed using field-direction callouts.
2. Repeat until clean, then increase speed.
3. Rotate toward the next action while traveling.
4. Run mirrored routes from both alliance sides.
5. Change starting heading while keeping route directions the same.

Penalty	Added time
Cone or field contact	+2 seconds

Missed stopping or scoring position	+3 seconds
Wrong field direction or major correction	+2 seconds
Unnecessary heading reset	+2 seconds

1:43–1:55 — Heading Recovery Drill

1. Move to open space and disable the robot.
2. Use the team's approved method to create an intentionally incorrect heading reference.
3. Enable; identify the mismatch with one small test input.
4. Stop safely, use the standard reset orientation, and reset heading.
5. Confirm all four field directions before normal driving.

Never surprise the driver with a heading-reference change while the robot is moving near people or structures.

Success target: safely correct the mismatch in under 15 seconds without collision.

1:55–2:00 — Timed Challenge and Review

Run one three-minute match-style challenge with objectives rather than a memorized path. Use normal match communication. Record cycles, collisions, missed alignments, wrong-direction starts, and resets.

Each driver then names one improvement, one recurring mistake, and one exercise to repeat. Record the results as the baseline.

Coaching Priorities

1. **Trust field directions:** drive toward the destination without converting through robot orientation.
2. **Separate the sticks mentally:** one commands field translation; the other commands rotation.
3. **Rotate with purpose:** select the heading needed for the next mechanism action.
4. **Use small test inputs after doubt:** confirm the reference before accelerating.
5. **Reset deliberately:** stop in open space, use the known orientation, reset, and verify.
6. **Accuracy before speed:** increase speed only after paths and headings are repeatable.

Practice Scorecard

Metric	Baseline	Best	Next target
Direction calls correct in a row			
Clean fixed-heading shapes			
Target alignments out of 10			
Fastest clean full-field route			
Cycles in timed challenge			
Wrong-direction starts			
Heading-recovery time			